

UNIVERSITY COLLEGE OF ENGINEERING

End Semester Examination

Computer Organization and Architecture (BCS302)

Roll No. _____

Subject: Computer Organization and
Architecture

Time: 3 Hours

Max. Marks: 70

Code: BCS302

General Instructions:

1. Attempt all questions in Section A and Section C.
2. Attempt any three questions from Section B.
3. Scientific calculators are permitted.
4. Assume suitable data where necessary.

SECTION A — SECTION A

Attempt all questions

2 marks each | Attempt all questions

Q.No.	Question	Marks	CO
1	In a daisy-chaining bus arbitration scheme, explain why the physical distance of a device from the bus controller creates a 'priority-starvation' scenario for downstream devices.	2	CO1
2	The IEEE 754 standard uses a biased exponent rather than a standard 2's complement representation. Justify this design choice in the context of floating-point comparison operations.	2	CO2
3	A processor features a 4-stage pipeline. If a sequence of instructions contains heavy data dependencies, analyze why the actual speedup might drop significantly below the theoretical maximum of 4.	2	CO3
4	A software loop repeatedly accesses memory locations that map to the same set in a direct-mapped cache, causing 'thrashing'. Propose how changing the mapping to 2-way set associative would mitigate this.	2	CO4
5	During a high-speed data transfer from a disk drive, the CPU is frequently interrupted, leading to system lag. Explain how a Direct Memory Access (DMA) controller solves this by 'stealing' bus cycles.	2	CO5
6	Compare a stack-organized CPU with a general-purpose register-organized CPU. Which one would typically require fewer bits to represent an 'ADD' instruction, and why?	2	CO1
7	In Booth's Multiplication Algorithm, if the current multiplier bit is 1 and the previous bit was 0 (bit pair 10), what specific arithmetic operation is performed on the partial product, and what is the logic behind it?	2	CO2

SECTION B — SECTION B*Attempt any 3 questions**7 marks each | Attempt any 3*

Q.No.	Question	Marks	CO
8	Design a 16-bit instruction format for a hypothetical processor that supports 32 general-purpose registers and 15 distinct operation codes. If the processor must support an 'Immediate' addressing mode for constants up to 255, evaluate if a single-word instruction is sufficient. Suggest an alternative if not.	7	CO3
9	A digital system employs a 2-level memory hierarchy with a main memory of 1 MB and a cache of 64 KB. If the cache uses 4-way set-associative mapping with a block size of 128 bytes, calculate the number of bits in the Tag, Set, and Word fields. Explain the 'Replacement Policy' conflict that arises when all lines in a set are full.	7	CO4
10	A real-time embedded system manages three I/O devices with varying urgencies. Describe the hardware implementation of a 'Priority Interrupt Controller' using a priority encoder. How does the CPU distinguish between a software exception and an external hardware interrupt during the fetch-execute cycle?	7	CO5
11	Analyze the architectural differences between a Synchronous and an Asynchronous bus. In a scenario where a fast CPU needs to communicate with a very slow legacy peripheral, evaluate which bus type would be more reliable and explain the 'Handshaking' protocol involved.	7	CO1
12	Perform the multiplication of binary numbers (-9) and (+6) using Booth's Algorithm, showing every step of the register contents (AC, QR, Qn+1). Explain why Booth's algorithm is considered superior to the hardware-based 'Shift and Add' method for signed operands.	7	CO2

SECTION C — SECTION C

Attempt all questions. Choose one option from each question.

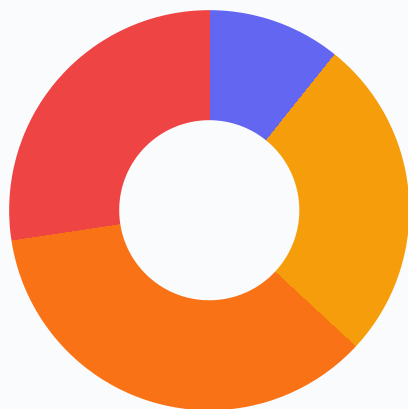
7 marks each | Attempt all questions

Q.No.	Question	Marks	CO
13	(a) A pipelined processor encounters a 'Branch' instruction at stage 2. Evaluate the performance penalty if the pipeline is flushed vs. if 'Branch Prediction' is used. Describe the hardware requirements for a simple 1-bit dynamic branch predictor.	7	CO3
	<i>OR</i>		
	(b) Contrast the design of a Hardwired Control Unit with a Microprogrammed Control Unit. For a high-performance RISC processor with a limited instruction set, justify which control unit design would be more appropriate for maximizing clock speed.	7	CO3
14	(a) Design a 2D memory organization for a 4096 x 16 bit RAM. Explain the 'Selection Logic' and why a 2 1/2D organization might be preferred to reduce the complexity of the row and column decoders.	7	CO4
	<i>OR</i>		
	(b) In a virtual memory system, a 'Page Fault' occurs. Detail the exact sequence of events the OS and hardware perform to resolve this. Analyze how a Translation Lookaside Buffer (TLB) minimizes the performance overhead of address translation.	7	CO4
15	(a) Compare 'Programmed I/O' with 'Interrupt-Initiated I/O'. In a system where the CPU is 1000 times faster than the I/O device, calculate the percentage of CPU time wasted in a polling loop and justify the transition to an interrupt-driven model.	7	CO5
	<i>OR</i>		
	(b) Explain the concept of 'I/O Channels'. Differentiate between a Selector Channel and a Multiplexer Channel. Which one would you implement for a system managing multiple low-speed terminals versus a single high-speed magnetic tape drive?	7	CO5
16	(a) Identify the role of 'Addressing Modes' in improving code density. For a task involving the traversal of a large 2D array, justify why 'Base-Register' plus 'Indexed' addressing is superior to absolute addressing.	7	CO1
	<i>OR</i>		
	(b) A system uses a centralized bus arbiter with a 'Rotating Priority' (Round Robin) scheme. Analyze how this prevents device starvation compared to a 'Fixed Priority' scheme. Illustrate the control signals (Bus Request, Bus Grant) involved in one arbitration cycle.	7	CO1

Q.No.	Question	Marks	CO
17	(a) Design a 4-bit Look-Ahead Carry Adder (LACA) using Carry Generate (G) and Carry Propagate (P) functions. Mathematically demonstrate why the propagation delay of a LACA is independent of the number of bits compared to a Ripple Carry Adder.	7	CO2
	<i>OR</i>		
	(b) Analyze the IEEE 754 Single Precision format for representing floating-point numbers. Convert the decimal value -25.75 into its 32-bit hexadecimal equivalent. Discuss how 'Underflow' is handled using subnormal numbers.	7	CO2

Question Paper Analysis

Bloom's Taxonomy Distribution



- K1 – Remember: 0m (0%)
- K2 – Understand: 0m (0%)
- K3 – Apply: 9m (11%)
- K4 – Analyze: 22m (26%)
- K5 – Evaluate: 30m (36%)
- K6 – Create: 23m (27%)

Course Outcome Distribution

